Schumpeterian Growth Model with Many Intermediates*



"What Descartes did was a good step. You have added much several ways. […]
If I have seen further it is by standing on the shoulder of Giants"
Isaac Newton to Robert Hooke, Feb 5 1675.

Intermediate Goods (for each j)

Final Good Firms
- Production:
$$Y \equiv L_Y^{1-\alpha} J^{\alpha} \left(\prod_{j=1}^{J} a_j^{1-\alpha} x_j^{\alpha} \right)^{1/J}$$
- J is fixed
- Spends $p_j x_j = \frac{\alpha}{J} Y$ in each j.
- Take prices as given.
- Zero profits:
$$Y - wL_Y - \sum_{j=1}^{J} p_j x_j = 0$$

Leader
- Quality $a(N_j) = (1+\gamma)^{N_j}$
- Price $p = (1+\gamma)r$
- Profits: $\pi = \frac{\gamma}{1+\gamma}\frac{\alpha}{J}Y$

Temporary monopoly:

Future Leader, $a(N_j+1) = (1+\gamma)^{N_j+1}$

Limits price: $p_{leader} \leq (1+\gamma) p_{fringe}$

Fringe Firm
- Quality $a(N_j - 1)$
- Marginal cost: r
- Cannot produce with profits if:
$$\frac{a(N_j)}{p(N_j)} \geq \frac{a(N_j-1)}{p(N_j-1)}$$

Future Fringe Firm - Current Leader, $a(N_j)$

Creative Destruction

Leader Replacement with prob $E[dN]$

R&D Sector
- Innovation: dN
  Improves industrial quality with some probability.
- Uses labor: L_R
- Free entry condition:
$$V \equiv \frac{\pi}{r - g_\pi + \mathrm{E[dN]}} = w\frac{1}{J}\frac{L_R}{E[dN]} \equiv F$$

Households
- Consume final output Y:
- Supply $L = L_Y + L_R$
- Receives wages wL.
- Earn all profits.

Payments for x_j

Int. Goods x_j

Profits π

Rent rK_j

Labor L_y

Wages wL_Y

Final output Y

Research Labor L_R

Wages wL_R

Mechanism

- ▶ Market Power (temporary)
  - ▶ Profits
    - ▶ R&D Incentives
      - ▶ Innovation (E[dN])
        - ▶ Creative Destruction
          - ▶ New Leader

Aggregate Outcomes

- Intermediate production: $\forall j; x_j = \frac{K}{J}$
- Aggregate Supply: $Y = K^{\alpha}(AL_Y)^{1-\alpha}$
- Technology: $A^{1-\alpha} = \left(\prod_{j=1}^{J} a(N_j)^{1-\alpha}\right)^{1/J}$
- S_R determined by:
$$\frac{s_R}{(1-s_R)} = \frac{\alpha\gamma/(1+\gamma)}{1-\alpha} \times \frac{E[dN]}{r - g_L + E[dN]}$$

Dynamic Relationships

- Profit grows endogenously in the same way as output:
$$g_\pi = g_Y = g_L + g_A$$
- Technology growth: $g_A = \gamma E[dN]$
- Effective discount rate: $r + E[dN]$

*Summary diagram prepared for Econ 414 – Winter 2026

Comments: bdaza@umich.edu This version: 2/20/26